NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-94/4J
11 September 1985
SUPERSEDING
MIL-R-94/4H
26 October 1981

PERFORMANCE SPECIFICATION

RESISTORS, VARIABLE, COMPOSITION, STYLE RV2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RV2 composition, variable, resistors.

2. APPLICABLE DOCUMENT

2.1 Government document.

2.1.1 Specification. Unless otherwise specified, the following specification of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-94 - Resistors, Variable, Composition, General Specification for.

(Copies of specification required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-94, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Mounting and locking nuts. The mounting nut shall be 0.093-inch thick and unless otherwise specified, shall measure .5 inch across the hexagonal flats. The locking nut shall be within the limits of .151 and 0.234-inch thick and shall measure .5 inch across the hexagonal flats. Thread size shall be 3/8-32NEF-2B.

3.2.2 Internal-tooth lockwasher. Internal-tooth lockwasher of suitable size shall be supplied.

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.047 inch.

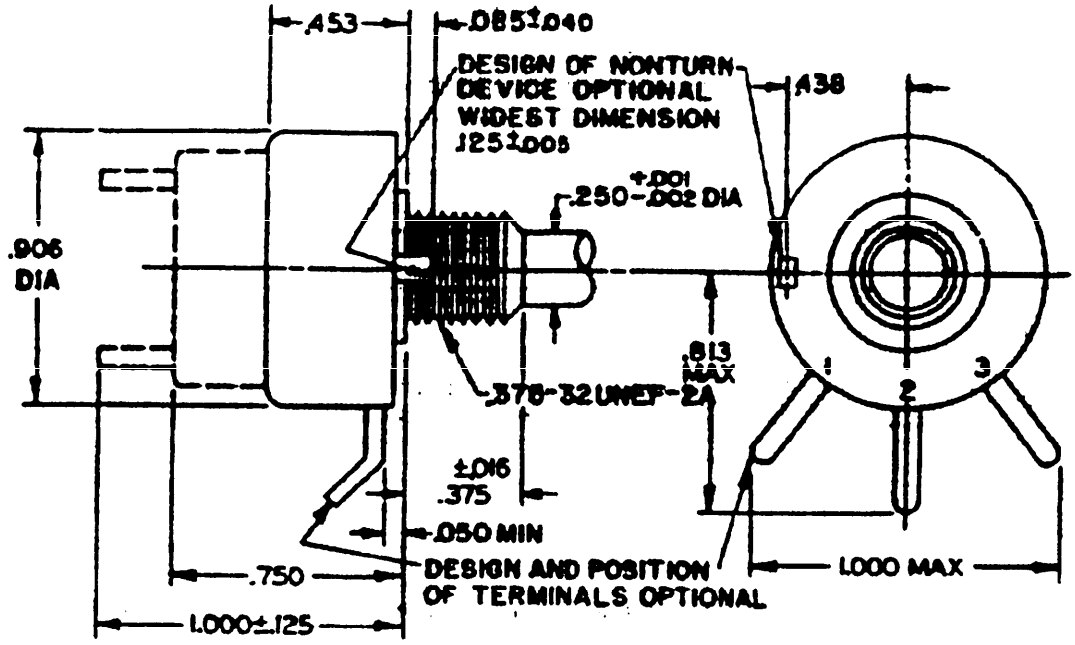
3.3 Total mechanical rotation. The total mechanical rotation without switch shall be within the limits of 266° and 338°; without switch, within the limits of 251° and 318°.

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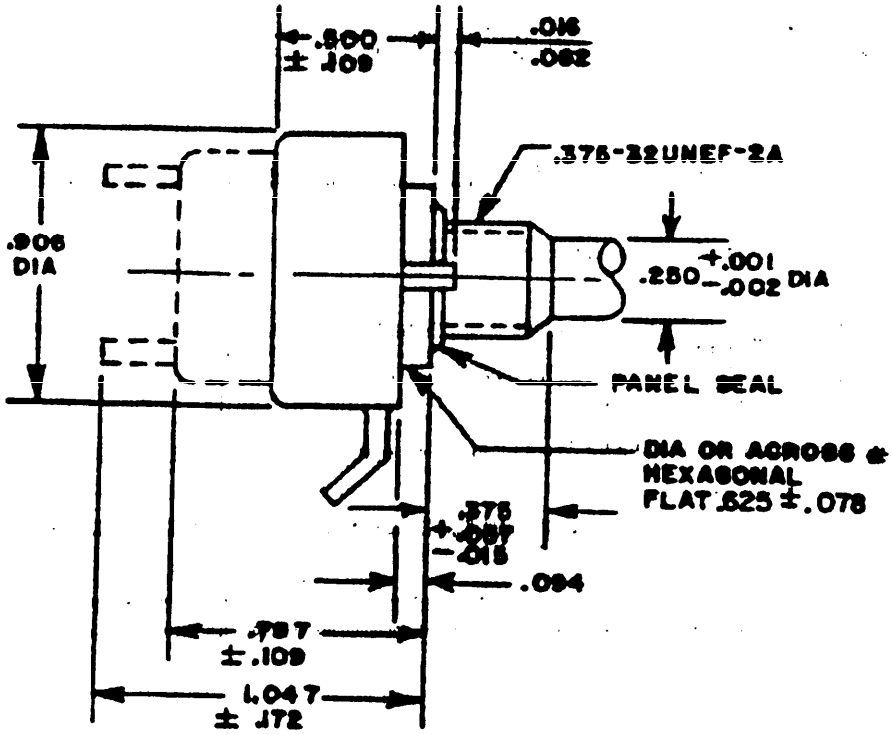
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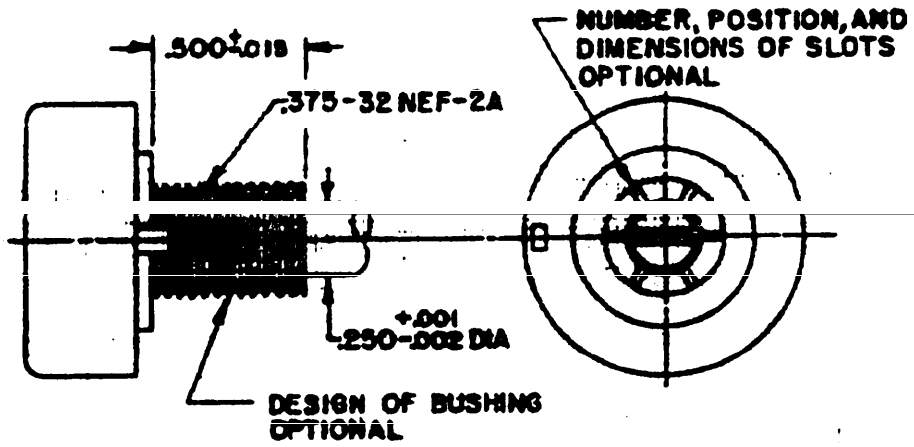
STANDARD-BUSHING TYPE



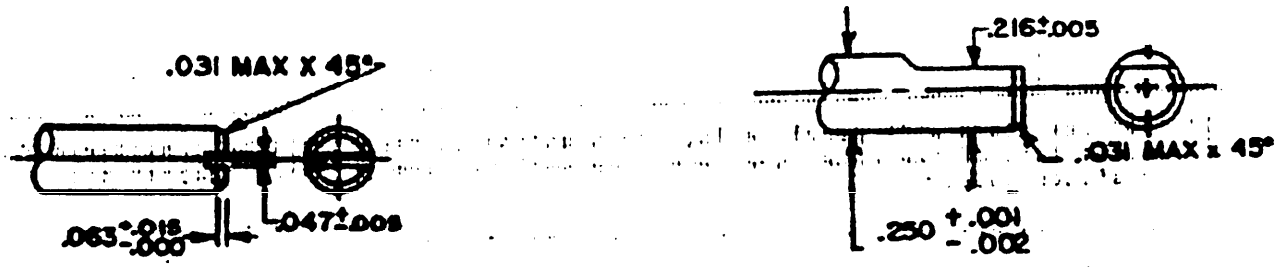
SHAFT-AND PANEL-SEALED BUSHING TYPE

FIGURE 1. Style RV2 resistors.

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LOCKING-BUSHING TYPE



SLOTTED SHAFT

FLATTED SHAFT

INCHES	MM	INCHES	MM
.001	0.03	.125	3.18
.002	0.05	.172	4.37
.005	0.13	.216	5.49
.015	0.38	.250	6.35
.016	0.41	.375	9.53
.031	0.79	.438	11.13
.040	1.02	.463	11.81
.045	1.14	.500	12.70
.047	1.19	.625	15.88
.050	1.27	.687	17.70
.057	1.45	.750	19.05
.062	1.57	.797	20.24
.063	1.60	.813	20.65
.085	2.16	.906	23.01
.094	2.39	1.000	25.40
.109	2.77	1.047	26.59

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 1. Style RV2 resistors - Continued.

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3.4 Shaft length. The length of operating shaft shall be in accordance with table I.

TABLE I. Length of operating shaft. 1/

Symbol	Length of operating shaft from mounting surface of resistor (nominal)		
	Knurled	Slotted	
	Bushings H and S (inches, ±1/32)	Bushings H and S (inches, ±1/32)	Bushings L and T (inches, ±1/32)
A	---	---	5/8
B	---	1 1/2	---
D	7/8	7/8	7/8 2/
E	---	---	---
G	---	1 1/4 2/	---
J	---	2 2/	---
K	---	2 1/2 2/	---

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

2/ For replacement purposes only; not for new design.

3.5 Nominal total resistance and rated continuous working voltage. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F		Taper A	Tapers C and F
Ohms			Ohms			Megohms		
100	10	7	5,000	59	42	.10	316	200
150	12	9	6,000	71	50	.15	350	200
200 2/	14	10	7,500	87	62	.20 2/	350	200
250	16	11	10,000	100	71	.25	350	200
350	19	13	15,000	123	87	.35	350	200
500	22	16	20,000 2/	140	100	.50	350	200
750	27	19	25,000	158	112	.75	350	200
1,000	32	24	35,000	187	132	1.0	350	200
1,500	39	27	50,000	224	158	1.5	350	200
2,000 2/	44	31	75,000	274	194	2.0	350	200
2,500	50	35				2.5	350	200

1/ Rated continuous working voltage at 70°C.

2/ For replacement purposes only. Not for new design.

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3.6 Power rating. The power rating shall be 1 watt for taper-A resistors and 1/2 watt for tapers-C and -F resistors.

3.7 Torque.

3.7.1 Operating. The torque required to effect rotation shall be not less than 1 ounce-inch nor more than 6 ounce-inches.

3.7.2 Stop. The torque applied through the operating shaft to the stops shall be 8 pound-inches.

3.7.3 Locking bushing. For the locking-bushing type, the locking nut shall be tightened with a torque of 10 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 30 ounce-inches is applied to the shaft.

3.9 Thermal cycling. The resistors shall not change in resistance in excess of 6 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-94, and as specified herein.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential at reduced barometric pressure shall be 450 volts, rms.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-94.

6. NOTES

6.1 Notes. The notes specified in MIL-R-94 are applicable to this specification.

6.2 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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